2201NSC Linear Algebra and Applications

Week 4

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Linear Transformations

Linearity, images, matrices and recurrence models

A transformation maps elements between sets

A transformation maps elements of one set to elements of another set. Transformations can act on many kinds of mathematical objects.
In this chapter, we focus on transformations on vectors, particularly

$$T : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

where an input vector from the **domain** $\mathbb{R}^n$ is mapped to an output vector in the **codomain** $\mathbb{R}^m$.

Example

$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} x_3 \\ x_1 + x_2 \end{pmatrix}.$$

The output $T(\mathbf{x})$ is called the image of $\mathbf{x}$

If $T : \mathbb{R}^n \to \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$, then

$$\boxed{T(\mathbf{x}) \text{ is the } \textbf{image} \text{ of } \mathbf{x} \text{ under } T.}$$

Find an image

For

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + 2x_2 \\ 2x_1 - x_2 \end{pmatrix},$$

the image of $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is

$$T\left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 7 \\ -1 \end{pmatrix}.$$

A linear transformation preserves two operations

A transformation T is **linear** when, for all vectors $\mathbf{u}, \mathbf{v}$ in its domain and every scalar c,

$$\boxed{T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})}$$

$$\boxed{T(c\mathbf{u}) = cT(\mathbf{u}).}$$

Additivity

Transforming a sum gives the sum of the images.

Scalar multiplication

Scaling before or after transforming gives the same result.

To prove linearity, verify both conditions

Let

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

$$\begin{aligned} T(\mathbf{u}+\mathbf{v}) &= T\left(\begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}\right) \\ &= \begin{pmatrix} u_2 + v_2 \\ u_1 + v_1 \end{pmatrix} \\ &= \begin{pmatrix} u_2 \\ u_1 \end{pmatrix} + \begin{pmatrix} v_2 \\ v_1 \end{pmatrix} \\ &= T(\mathbf{u}) + T(\mathbf{v}), \\ T(c\mathbf{u}) &= T\left(\begin{pmatrix} cu_1 \\ cu_2 \end{pmatrix}\right) \\ &= \begin{pmatrix} cu_2 \\ cu_1 \end{pmatrix} = c\begin{pmatrix} u_2 \\ u_1 \end{pmatrix} = cT(\mathbf{u}). \end{aligned}$$

Therefore T is linear.

To disprove linearity, one counterexample is enough

Consider

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 x_2 \\ 3 \end{pmatrix}.$$

Take

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then

$$T(\mathbf{u}+\mathbf{v}) = T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 3 \end{pmatrix},$$

$$T(\mathbf{u}) + T(\mathbf{v}) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}.$$

Conclusion

Additivity fails for this pair, so T is not linear.

Common warning signs of a nonlinear mapping

Products	Added constants	Fixed output
$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 x_2 \\ 2x_1 \end{pmatrix}$ Input components are multiplied.	$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ x_1 - 1 \end{pmatrix}$ The output contains a constant shift.	$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + x_2 \\ 2 \end{pmatrix}$ One component is always nonzero.

Use these as clues

A counterexample must still identify a failed linearity condition.

Linear transformations can change dimension

$\mathbb{R}^2 \to \mathbb{R}^3$

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 \\ x_2 \\ x_1 + 2x_2 \end{pmatrix}.$$

Two input coordinates produce three output coordinates.

$\mathbb{R}^3 \to \mathbb{R}$

$$T\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = x_1 + x_3.$$

The image is a scalar, regarded as a vector in $\mathbb{R}$.

Changing dimension does not prevent a transformation from being linear.

Recognise common transformations in the plane

Rotations

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} -y \\ x \end{pmatrix}$$

90° anticlockwise

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} -x \\ -y \end{pmatrix}$$

180° about the origin

Reflections

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} -x \\ y \end{pmatrix}$$

Reflection in the y-axis

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ -y \end{pmatrix}$$

Reflection in the x-axis

Uniform contraction

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \frac{1}{2}\begin{pmatrix} x \\ y \end{pmatrix}$$ scales every vector by $\frac{1}{2}$.

Every linear transformation has a standard matrix

For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, there is an $m \times n$ matrix A such that

$$\boxed{T(\mathbf{x}) = A\mathbf{x}.}$$

Dimensions

$$T : \mathbb{R}^3 \to \mathbb{R}^2 \quad \Longrightarrow \quad A \text{ has size } 2 \times 3.$$

The number of columns matches the domain dimension; the number of rows matches the codomain dimension.

The columns of A are images of standard basis vectors

For $T : \mathbb{R}^2 \to \mathbb{R}^2$,

$$A = \begin{pmatrix} | & | \\ T(\mathbf{e}_1) & T(\mathbf{e}_2) \\ | & | \end{pmatrix}.$$

Example: clockwise rotation

If

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} y \\ -x \end{pmatrix},$$

then

$$T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

A $\mathbb{R}^3 \to \mathbb{R}^2$ standard matrix has three columns

Let

$$T\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} x_3 \\ x_1 + x_2 \end{pmatrix}.$$

The images of the standard basis vectors are

$$T\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad T\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad T\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Therefore,

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Composition reverses the order of the matrices

Suppose

$$T_1(\mathbf{x}) = A_1\mathbf{x}, \qquad T_2(\mathbf{x}) = A_2\mathbf{x}.$$

Applying T_1 first and then T_2 gives

$$\begin{aligned} T_2(T_1(\mathbf{x})) &= A_2(A_1\mathbf{x}) \\ &= (A_2A_1)\mathbf{x}. \end{aligned}$$

Therefore, the transformation matrix for T_1 followed by T_2 is A_2A_1.

Remember the order

The transformation applied **first** appears on the **right** of the matrix product.

Matrix powers represent repeated transformations

If $T(\mathbf{x}) = A\mathbf{x}$, then applying T repeatedly gives

$$T(T(\mathbf{x})) = A^2\mathbf{x}, \qquad T^k(\mathbf{x}) = A^k\mathbf{x}.$$

Why this matters

Many recurrence relations update a small state vector in the same way at every step. The update is a linear transformation, so matrix powers can jump directly to a later state.

The Fibonacci recurrence is a linear transformation

Since $F_n = F_{n-1} + F_{n-2}$,

$$\begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} = T\left(\begin{pmatrix} F_{n-1} \\ F_{n-2} \end{pmatrix}\right) = \begin{pmatrix} F_{n-1} + F_{n-2} \\ F_{n-1} \end{pmatrix}.$$

Thus

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + x_2 \\ x_1 \end{pmatrix}.$$

The images of the standard basis vectors are

$$T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

These images form the columns of A, so

$$\boxed{A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.}$$

Use the Fibonacci matrix with the correct power

Starting from

$$\begin{pmatrix} F_1 \\ F_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

eleven updates are needed to reach F_{12}:

$$\begin{pmatrix} F_{12} \\ F_{11} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{11} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 144 \\ 89 \end{pmatrix}.$$

Therefore,

$$\boxed{F_{12} = 144.}$$

Short Revision Quiz

Linearity, images, standard matrices and recurrences

Question 1 — Conditions for linearity

Which pair of conditions defines a linear transformation?

A. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$

B. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})T(\mathbf{v})$ and $T(c\mathbf{u}) = T(\mathbf{u})$

C. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + \mathbf{v}$ and $T(c\mathbf{u}) = c + T(\mathbf{u})$

Question 2 — Terminology

For a transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ and an input $\mathbf{x} \in \mathbb{R}^n$, what is $T(\mathbf{x})$ called?

A. The domain of $\mathbf{x}$

B. The image of $\mathbf{x}$ under T

C. The standard basis of T

Question 3 — Find an image

Let

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}.$$

Find the image of $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ under T.

A. $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

C. $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

D. $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

Question 4 — Using additivity

If T is linear, which expression is equal to $T(\mathbf{u}+\mathbf{v})$?

A. $T(\mathbf{u})+T(\mathbf{v})$

B. $T(\mathbf{u})T(\mathbf{v})$

C. $T(\mathbf{u})+\mathbf{v}$

Question 5 — Matrix dimensions

If $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear and $T(\mathbf{x}) = A\mathbf{x}$, what are the dimensions of A?

A. 2×3

B. 3×2

C. 3×3

D. 2×2

Question 6 — Build a standard matrix

Suppose

$$T\left(\begin{pmatrix}1\\0\end{pmatrix}\right)=\begin{pmatrix}2\\-1\end{pmatrix},\qquad T\left(\begin{pmatrix}0\\1\end{pmatrix}\right)=\begin{pmatrix}3\\4\end{pmatrix}.$$

Find the standard matrix A such that

$$T(\mathbf{x})=A\mathbf{x}.$$

A. $\begin{pmatrix}2 & -1\\3 & 4\end{pmatrix}$ **B.** $\begin{pmatrix}2 & 3\\-1 & 4\end{pmatrix}$ **C.** $\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$

Question 7 — Read a transformation from its matrix

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Given $T(\mathbf{x}) = A\mathbf{x}$, complete

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = ?$$

A. $\begin{pmatrix} x_1 + 2x_2 \\ -x_2 \end{pmatrix}$ **B.** $\begin{pmatrix} x_1 \\ 2x_1 - x_2 \end{pmatrix}$ **C.** $\begin{pmatrix} x_1 + x_2 \\ 2x_1 - x_2 \end{pmatrix}$

Question 8 — True or false

Every linear transformation must map $\mathbb{R}^n$ back into the same space $\mathbb{R}^n$.

A. True

B. False

Question 9 — Columns of a standard matrix

Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be linear. Which vectors form the columns of its standard matrix?

A. The images $T\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $T\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$

B. The vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$

C. Any two vectors in the codomain

Question 10 — Composition of transformations

Suppose

$$T_1(\mathbf{x}) = A_1\mathbf{x}, \qquad T_2(\mathbf{x}) = A_2\mathbf{x}.$$

Which matrix represents applying T_1 first, followed by T_2?

A. $A_1 A_2$

B. $A_2 A_1$

C. $A_1 + A_2$

Quiz answers 1–5

1 — A

$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$.

2 — B

$T(\mathbf{x})$ is the image of $\mathbf{x}$ under the transformation T.

3 — A

The image is $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

4 — A

Linearity gives
$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.

5 — A

A is 2×3.

Quiz answers 6–10

6 — B

$$A = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}.$$

The given images form its columns.

7 — A

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1 + 2x_2 \\ -x_2 \end{pmatrix}.$$

8 — B

False: linear transformations may map between spaces of different dimensions.

9 — A 10 — B

The columns are the images of the standard basis vectors. Applying T_1 first and T_2 second gives $T_2(T_1(\mathbf{x})) = (A_2A_1)\mathbf{x}$.

Final takeaway

Linearity preserves sums and scalar multiples; the standard matrix records the images of the standard basis vectors.